1. Air Is Stuff

Crush a can with nothing

It has mass, it pushes, and it pushes hard

Lesson 1 · about 60 minutes · everyone — kitchen care only

What you need

- Empty aluminium drink cans
- Tongs, a hob, a bowl of iced water
- A boiled egg, peeled, and a glass bottle it will not quite fit into
- A drinking glass, card, water
- A balloon and a plastic bottle
- Bathroom scales, a bicycle pump with a gauge

Predict first

Put a tablespoon of water in an empty can, boil it until steam pours out, then turn the can upside down into iced water. What happens, and *how fast*? Write both down. Then: how much does the air in the room weigh?

Do it

1. **Crush the can.** A tablespoon of water in the can, on the hob until steam is streaming out for a good ten seconds. Then, with tongs, invert it into iced water. It collapses instantly and loudly. Nothing touched it.
2. **The glass of water.** Fill a glass to the brim, lay a card over it, invert. The card stays put and the water does not fall out. Air is pushing *up* on it.
3. **Turn it sideways.** Rotate the inverted glass slowly. The card stays on at every angle, so the push is not gravity and it is not downward — pressure acts in every direction at once.
4. **The egg in the bottle.** Drop a burning strip of paper into the bottle and sit the peeled egg on the neck. As the flame goes out and the gas cools, the egg is pushed in.
5. **Weigh the push.** Pump a bicycle tyre to a known pressure and read the gauge. Then work out the force on a 3 in disc at that pressure — the same disc as your launcher's chamber end.

What it means

Air has mass	Roughly 1.2 kg per cubic metre. The air in an ordinary bedroom weighs about as much as a small child — something like 50 kg. It is easy to forget because you are floating in it.
And therefore pressure	The column of atmosphere above you presses at about 14.7 psi, or 101 kPa — roughly 10 tonnes on a square metre. You do not notice because it presses equally on every side of you.
Nothing sucks	The can was not sucked in. Steam drove the air out, then condensed to a few drops of water, leaving very little inside — and the ordinary outside air pressure, no longer balanced, crushed it. There is no such thing as suction; there is only pressure on one side that is not matched on the other.
This is the whole launcher	The launcher does the same thing in reverse: it raises the pressure *inside* until the imbalance drives a potato out. Both are the same idea.
The number that matters	On a 3 in bore, one atmosphere of *extra* pressure is about 100 lb of force. That is why a modest pressure moves a heavy potato fast.

Break it

More water	Try a whole cup in the can instead of a spoonful. Does it crush harder, or not at all? Why might too much water be worse?
No ice	Invert into room-temperature water. The effect is weaker. What does that tell you about how fast the steam needs to condense?
A steel can	Try a food tin. Nothing happens. The force is the same — so what changed?
Do the arithmetic	Area of a 3 in circle is about 7 in^2. At 14.7 psi that is over 100 lb of force from ordinary air. Now imagine three atmospheres.

The link to the launcher

Half the fuel in your launcher is already in the chamber and it is free: the oxygen in the air. Lesson 2 is about how much of it there is and why that sets exactly how much hairspray you may use. The pressure that ordinary air already exerts is the baseline that everything in Lesson 3 is measured against.